PyPSA-Libya: An Open-Source Energy System Model for National Decarbonization and Hydrogen Export Pathways

Mohamed Abuella[a]

[a]*Independent Researcher, Newcastle upon Tyne, UK*

Abstract

Libya possesses significant renewable energy potential that remains largely untapped. This paper presents PyPSA-Libya, a national-scale model built on the PyPSA-Earth framework, designed to explore optimal transition pathways for the Libyan power sector. We present two primary studies: (1) an electricity-only capacity expansion analysis comparing a zero-emission constraint against an unconstrained baseline at 36-cluster resolution, and (2) a sector-coupled hydrogen export study evaluating the levelized cost of exporting 70 TWh/year of hydrogen to Europe under three technological scenarios. Results indicate that while solar PV dominates the capacity mix due to favorable 2030 cost assumptions, the integration of hydrogen production significantly alters the grid's storage requirements. The model, data, and configuration files are released under an open-source license to support transparent energy planning in Libya.

Keywords: Energy system modelling, Libya, Green Hydrogen, Decarbonization, PyPSA-Earth

1. Introduction

The global imperative to mitigate climate change has accelerated the transition from carbon-intensive fossil fuels to renewable energy sources, in line with the objectives of the Paris Agreement [1, 2]. Petro-state economies face a distinct dual challenge during this transition: they must decarbonize their domestic power sectors while simultaneously diversifying their economies away from oil and gas dependence [1, 3]. Libya represents a critical case study of this paradigm. The country's political economy relies almost entirely on conventional fossil fuels, with the oil sector contributing approximately 95% of export earnings and 65% of Gross Domestic

Email address: mhdabuella@gmail.com (Mohamed Abuella)

Product (GDP) [1]. Over the past decade, political instability and administrative division have severely affected Libya's energy sector. Armed conflicts and unilateral shutdowns of major oil fields (such as those in early 2020 causing financial losses exceeding $9 billion) have repeatedly undermined the reliability of fuel supplies for electricity generation [1].

Concurrently, Libya's domestic power sector, monopolized by the state-owned General Electricity Company of Libya (GECOL), is facing a severe operational crisis [1]. The power grid relies completely on fossil fuels, with natural gas driving approximately 63% of generation and oil-fired plants accounting for 29% [1]. Rapid growth in energy demand, driven by rising temperatures and population growth, has led to a persistent peak-load electricity deficit of approximately 1,200 MW, resulting in frequent and widespread blackouts [1]. Furthermore, Libya's environmental footprint is substantial; in 2018, its per capita CO2 emissions amounted to 8.73 tonnes, surpassing the Chinese national average of 7.95 tonnes [1]. Although the Renewable Energy Authority of Libya (REAoL) set an ambitious target in 2010 to integrate a 10% renewable share (equivalent to 2,219 MW) into the national mix by 2025, this initiative was completely derailed by the civil conflict that erupted in 2011 [1].

Despite these challenges, Libya is exceptionally well-positioned to pivot toward a sustainable energy economy. The country possesses some of the highest solar and wind resource potentials in the world. The average global horizontal irradiation (GHI) over the Libyan territory yields an average potential solar photovoltaic (PV) productivity of 1.9 MWh/kW/year (with Al Kufra in the southeast reaching an optimal annual productivity of 2,100 kWh/kWp), while onshore wind potentials reach up to 500 W/m^2 along its southernmost borders, coastal Darna, and the Western Mountain region [1]. Recognizing these vast resources, pioneering research in the late 1980s first proposed a solar-hydrogen energy system for Libya, emphasizing that converting solar power to hydrogen would not only extend the lifespan of fluid fossil fuel reserves as raw chemical feedstocks but also position Libya as a leading exporter of clean hydrogen to Europe in lieu of oil and gas [3].

To evaluate the modern technical and economic feasibility of this transition, we introduce `PyPSA-Libya`, a high-resolution, open-source energy system model of Libya's power and hydrogen network. Built as a country-specific extension of the global `PyPSA-Earth` framework [4], our model is divided into two primary, complementary studies designed to address both domestic and international planning questions:

1. **Electricity-Only Capacity Expansion (Part 1):** This study models Libya's power sector in isolation at a high spatial resolution of 36 clustered nodes. It compares a zero-CO2 emission constraint (Case 1) against an unconstrained fossil-heavy baseline (Case 2) under 2030 technology-cost assumptions, captur-

ing transmission bottlenecks and identifying the optimal mix of solar, wind, and storage needed for complete domestic grid decarbonization.

2. **Sector-Coupled Hydrogen Export (Part 2):** This study models the spatial and economic implications of exporting 70 TWh/year of renewable or low-carbon hydrogen from Libya to European markets via maritime shipping [4]. Formulated at a 12-cluster resolution to maintain computational tractability, we evaluate three technology scenarios representing strictly green electrolysis (Scenario A), electrolysis coupled with carbon-captured steam methane reforming (SMR CC) representing blue hydrogen (Scenario B), and an unconstrained scenario representing grey, blue, and green pathways (Scenario C) .

To ensure robustness, our model incorporates a `connectivity_upgrade` workflow to resolve topological gaps and isolated buses inherent in OpenStreetMap (OSM) grid data for Libya. We report the Levelized Cost of Electricity (LCOE) and apply a paired-run differencing methodology to isolate the Net Levelized Cost of Hydrogen (Net LCOH) directly attributable to the export obligation. The model code, parameters, and results are released under an open-source license to establish a transparent, reproducible benchmark for national decarbonization and cross-border energy partnerships.

2. Related Work

The design of the `PyPSA-Libya` model and its scenario configurations build upon three main pillars of literature: (i) localized studies on Libyan renewable energy and hydrogen potential, (ii) advancements in open-source energy system modeling and the PyPSA framework, and (iii) the techno-economics of global hydrogen trade pathways, including the strategic integration of underground storage.

2.1. Renewable Potential and Hydrogen in Libya

The assessment of Libya's transition to alternative energy resources began with the pioneering work of Eljrushi and Veziroglu [3], who developed one of the earliest system models for a solar-hydrogen economy. Their study mathematically simulated the dynamic interaction between population growth, energy consumption, and the rapid depletion of fluid fossil fuel reserves, warning that Libya's domestic oil and gas production would fail to meet national demand within three to four decades without a strategic shift to solar-hydrogen production [3]. Almaktar and Shaaban [1] updated these resource assessments by conducting a comprehensive GIS and meteorological evaluation of the country's solar, wind, biomass, and geothermal capacities, defining

them as non-rivalry energy alternatives (NREAs) essential for national security and geopolitical resilience.

While these macro-level studies established resource potential, localized techno-economic analyses of hydrogen deployment remain sparse. Rahil et al. [5] provided a focused study on off-peak electricity utilization to drive onsite forecourt water electrolysis in Darnah, evaluating the localized levelized cost of fuel-station hydrogen under 2015 and 2030 cost horizons. However, their work was restricted to localized refueling stations and did not capture the systemic transmission, spatial balancing, or sector-coupling dynamics of a macro-scale national network. Our model bridges this gap by representing the entire national transmission topology and co-optimizing domestic and export infrastructure.

2.2. Open-Source Energy Modeling and the PyPSA Framework

Traditional energy planning models, such as those utilizing TIMES [6], PLEXOS, or LUT [4], are often constrained by commercial license barriers or proprietary datasets, which limits transparency and public collaboration in developing countries [4, 7]. To overcome these limitations, the Python for Power System Analysis (PyPSA) library was introduced as a fully open-source framework for high-resolution spatial and temporal capacity expansion modeling [7]. While the European model (`PyPSA-Eur`) successfully pioneered the integration of transmission reinforcement, temporal storage, and multi-sector coupling [7], its continental scope necessitated the development of single-country forks to integrate detailed national datasets.

Single-country PyPSA models have increasingly been used to bridge the gap between continental optimization and national policy planning. For example, `PyPSA-Spain` [7] extended the European framework by integrating high-resolution regional demand disaggregation and localized wind and solar hourly capacity factor calibrations. Furthermore, to address cross-border trade without the computational burden of co-optimizing the entire continent, `PyPSA-Spain` introduced a nested modeling approach utilizing precomputed time-dependent electricity prices from neighboring France and Portugal [7].

To expand open-source modeling beyond Europe, the `PyPSA-Earth` initiative was established, enabling the automated retrieval of global open-data (including OpenStreetMap grid data and ERA5 weather reanalysis) to generate validated power system models for any region on Earth [4]. Most recently, Abdel-Khalek et al. [4] developed the sector-coupled version of `PyPSA-Earth`, which incorporates multiple energy carriers (electricity, hydrogen, natural gas, biomass, oil, and heat) and major demand sectors (industry, transport, residential). Showcase applications of this model evaluated the repurposing of natural gas networks for hydrogen in Brazil and

calculated the infrastructure expansion needed for Egypt to meet 16% of European hydrogen demand by 2030, establishing the methodological foundation for our sector-coupled export study [4].

2.3. Techno-Economics of Global Hydrogen Trade and Geological Storage

The prospective scale of global hydrogen trade is subject to significant techno-economic and infrastructural uncertainty [8]. Fakhreddine et al. [8] reviewed the current trade modeling landscape, categorizing approaches into energy system, supply chain, and techno-economic models, and noting that most models focus on solar-rich exporters such as Australia, South America, and North Africa. In the Euro-Mediterranean context, Fattahi et al. [6] evaluated green energy partnerships between Europe and the Middle East and North Africa (MENA) using the TIAM-ECN model. Their results indicated that North Africa represents the most cost-effective supplier of gaseous hydrogen via pipeline infrastructure, whereas liquefied shipping from the Middle East can offer complementary diversification, generating total system savings for Europe of up to $40 billion per year by 2050 [6]. Alanazi et al. [2] further confirmed that international trade could meet over 31% of global decarbonized hydrogen demand (reaching 234 Mt by 2050), identifying North Africa-to-Europe as one of the most prominent global trade routes. However, hydrogen's role in the wider energy transition is heavily constrained by high conversion and transport losses, making direct electrification more competitive for applications such as space heating and light transport, and prioritizing hydrogen for hard-to-abate heavy industry, aviation, and long-duration storage [9].

A major challenge for hydrogen exporters is managing the extreme volatility of renewable generation while meeting constant, non-interruptible export or industrial demands [10, 11]. Large-scale water electrolysis systems require immense system flexibility, which often necessitates significant over-sizing of renewable capacities or battery storage [11]. Achhammer et al. [10] addressed this challenge by integrating GIS-based geological salt cavern potentials into `PyPSA-Earth` to model the impact of underground hydrogen storage (UHS) in Tunisia. Their study demonstrated that geological storage is not merely a technical buffer but a critical strategic asset; UHS reduces the levelized cost of imported hydrogen by approximately 3.7%, dampens localized price volatility, and prevents the depletion of domestic resource security under strict export quotas, thereby transforming purely extractive trade dynamics into "fair partnerships" [10]. By incorporating both shipping logistics and underground H2 storage constraints, `PyPSA-Libya` provides a comprehensive evaluation of how these geological advantages could shape Libya's export resilience.

3. Methodology and Model Basis

3.1. Framework and Spatial Resolution

The model utilizes the `PyPSA-Earth` Snakemake workflow. We employ two spatial resolutions:

- **Electricity-Only:** 36 clustered nodes to capture transmission bottlenecks and regional resource variations.
- **Sector-Coupled:** 12 clustered nodes to maintain computational tractability for complex hydrogen-production LPs.

3.2. Data and Cost Assumptions

We utilize ERA5 reanalysis weather data for the year 2013, resampled to 3-hour or 6-hour intervals. Technology costs are calibrated to the **2030 horizon**. A critical modification in `PyPSA-Libya` is the `connectivity_upgrade` setting, which ensures that every node in the OSM-derived topology has at least one connection, mitigating the issue of isolated buses common in Libyan data.

3.3. Scenario Design

The study is divided into two distinct parts:

3.3.1. Part 1: Electricity-Only Decarbonization

We compare two cases to isolate the cost of climate ambition:

- **Case 1 (Zero-CO2):** A strict `co2_limit` of 0.0, forcing a 100% renewable system (Solar, Wind, Battery, and H2 storage units).
- **Case 2 (Baseline):** An unconstrained CO2 cap (10^9 tonnes) reflecting a cost-minimal fossil-heavy baseline.

3.3.2. Part 2: Sector-Coupled Hydrogen Export

Following the methodology of [4], we model the export of **70 TWh/year** of hydrogen via shipping. We evaluate three scenarios:

- **Scenario A:** Electrolysis only (Green) with annual renewable matching.
- **Scenario B:** Electrolysis + Blue H2 (SMR with Carbon Capture).
- **Scenario C:** Unconstrained (Electrolysis + SMR + SMR CC).

4. Results: Electricity-Only Baseline

4.1. Capacity Mix and System Cost

[Insert discussion here: Case 1 likely shows a massive buildout of Solar PV and Battery storage compared to Case 2's reliance on CCGT/OCGT.]

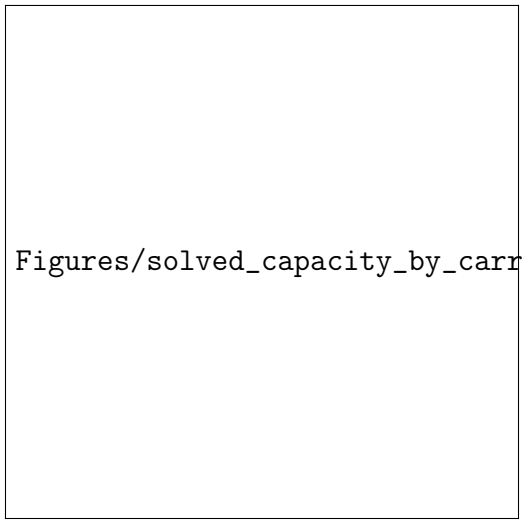


Figure 1: Optimal capacity mix for the 36-cluster electricity-only model (Zero-CO2 Case).

4.2. Transmission Bottlenecks

Utilizing the shadow prices (dual variables) of the transmission constraints, we identify key corridors requiring reinforcement to facilitate 100% renewable penetration.

5. Results: Sector-Coupled Hydrogen Export

5.1. Levelized Cost of Hydrogen (LCOH)

We report the **Net LCOH**, calculated by differencing the total system cost of an export run and a zero-export baseline.

5.2. Spatial Investment Patterns

The optimizer allocates hydrogen production near high-irradiance zones and coastal export hubs.



Figure 2: Map of transmission line loading and identified bottlenecks in the Libyan grid.

6. Discussion and Limitations

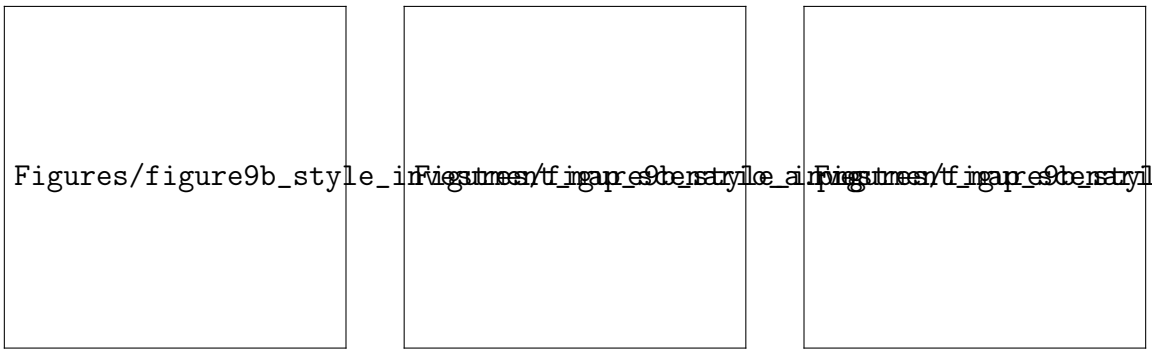
Key findings indicate that onshore wind is often priced out by Solar PV in the 2030 horizon despite Libya's strong wind resource. Furthermore, the reliance on OSM data remains a limitation; while `connectivity_upgrade` ensures a solved network, it may not perfectly reflect the operational reality of GECOL's transmission system. Future work should integrate verified geological data for underground H2 storage, which is currently modeled as functionally unconstrained.

7. Conclusion

PyPSA-Libya provides the first open-source, high-resolution look at Libya's energy future. The results suggest that Libya could become a competitive exporter of

Table 1: Summary of H2 Export Costs (2030 Assumptions)

Metric	Scenario A	Scenario B	Scenario C
Net LCOH [EUR/MWh]	[Value]	[Value]	[Value]
Net LCOH [EUR/kg LHV]	[Value]	[Value]	[Value]
System LCOE [EUR/MWh]	[Value]	[Value]	[Value]



(a) Scenario A (b) Scenario B (c) Scenario C

Figure 3: Spatial investment allocation for hydrogen production and supporting infrastructure.

green hydrogen to Europe, provided that transmission bottlenecks are addressed.

Code and Data availability

The model code is available at `https://github.com/mhdella/pypsa-libya`. Results and network files are archived on Zenodo at [DOI Placeholder].

References

[1] M. Almaktar, M. Shaaban, Prospects of renewable energy as a non-rivalry energy alternative in libya, Renewable and Sustainable Energy Reviews 143 (2021) 110852. doi:10.1016/j.rser.2021.110852.

[2] K. Alanazi, S. Mittal, A. Hawkes, N. Shah, Renewable hydrogen trade in a global decarbonised energy system, International Journal of Hydrogen Energy 101 (2025) 712–730. doi:10.1016/j.ijhydene.2024.12.452.

[3] G. S. Eljrushi, T. N. Veziroglu, Solar-hydrogen energy system for libya, International Journal of Hydrogen Energy 15 (12) (1990) 885–894. doi:10.1016/0360-3199(90)90077-C.

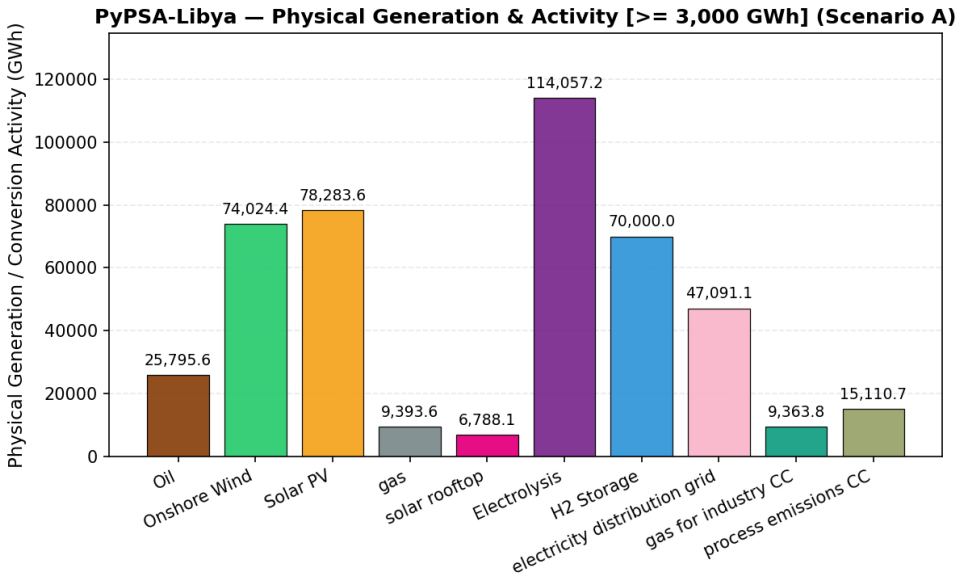


Figure 4: Capacity Scenario A

[4] H. Abdel-Khalek, L. Schumm, E. Jalbout, M. Parzen, C. Schauß, D. Fioriti, Pypsa-earth sector-coupled: A global open-source multi-energy system model showcased for hydrogen applications in countries of the global south, Applied Energy 383 (2025) 125316. doi:10.1016/j.apenergy.2025.125316.

[5] A. Rahil, R. Gammon, N. Brown, Techno-economic assessment of dispatchable hydrogen production by multiple electrolysers in libya, Journal of Energy Storage 16 (2018) 46–60. doi:10.1016/j.est.2017.12.016.

[6] A. Fattahi, F. D. Longa, B. van der Zwaan, Opportunities of hydrogen and ammonia trade between europe and mena, International Journal of Hydrogen Energy 83 (2024) 967–974. doi:10.1016/j.ijhydene.2024.08.021.

[7] C. Gallego-Castillo, M. Victoria, Pypsa-spain: An extension of pypsa-eur to model the spanish energy system, Energy Strategy Reviews 60 (2025) 101764. doi:10.1016/j.esr.2025.101764.

[8] J. Fakhreddine, P. E. Dodds, I. Butnar, Global hydrogen trade pathways: A review of modelling advances and challenges, International Journal of Hydrogen Energy 129 (2025) 236–252. doi:10.1016/j.ijhydene.2025.04.203.

[9] N. Johnson, M. Liebreich, D. M. Kammen, P. Ekins, R. McKenna, I. Staffell, Realistic roles for hydrogen in the future energy transition, Nature Reviews Clean Technology (2025). doi:10.1038/s44359-025-00050-4.

[10] A. Achhammer, D. Fioriti, A. Patonia, M. Sterner, The impact of hydrogen underground storage on export dynamics and system resilience: A gis-based integration of salt caverns into pypsa-earth, Energy 360 (2026) 141729. doi:10.1016/j.energy.2026.141729.

[11] T. Terlouw, C. Bauer, R. McKenna, M. Mazzotti, Large-scale hydrogen production via water electrolysis: a techno-economic and environmental assessment, Energy & Environmental Science 15 (2022) 3583–3602. doi:10.1039/d2ee01023b.